Dipendra Sharma

Aspiring Backend Engineer | Building Scalable, Reliable System

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Summary

I'm a backend-focused developer who enjoys building reliable systems and learning new technologies through real-world projects. I've built and deployed **REST APIs**, worked with relational databases, and I'm currently learning **Distributed System**. I have a strong foundation in **databases**, **data structures**, **algorithms**, **operating systems**, and object-oriented programming. I approach problems from first principles, tracing issues through system behavior. I take ownership of features end-to-end and enjoy working across the backend stack. I'm currently exploring **Redis**, caching strategies, system design, load balancing, and distributed systems to build scalable and maintainable software.

Technical Skills

Languages: JavaScript, TypeScript, Java, C++, SQL
Backend: Node.js, Go, Express.js, REST APIs, RBAC, Redis
Databases: PostgreSQL (indexing, CTEs, EXPLAIN ANALYZE, connection pooling), MongoDB, MySQL, Supabase
DevOps & Linux: Docker, AWS (EC2, S3, IAM), Git, GitHub Actions, SSH, Render, Linux, bash, cron, Apache Bench
Security & Performance: Rate limiting, bcrypt, JWT, query optimisation, load testing, input validation

Projects

Yeti Jobs | GitHub | Live **Jan 2026 – Apr 2026**

- Build **50+ REST APIs** with role-based access control (RBAC) across 3 user roles, backed by **10 normalized PostgreSQL tables**. Wrote complex SQL with JOINs, CTEs, and composite indexes, dropping query times by **30%** (7ms to 5ms) validated via EXPLAIN ANALYZE. Configured connection pooling for **20 concurrent DB connections**.
- Debugged **10+ production issues** to root cause, resolving PostgreSQL connection pool leaks and duplicate cron executions. Traced stack traces and analyzed query plans to permanently fix bottlenecks before they impacted users.
- Load-tested with Apache Bench (**1,000 requests** at **100 concurrent users** achieving **67 req/sec** with **0 failures**) and authored **28+ integration tests** using Jest and Supertest to ensure API reliability.
- Optimized backend deployment using multi-stage Docker builds (Alpine), reducing the image size by **74%**. Deployed to **AWS EC2** with Nginx as a reverse proxy, configured GitHub Actions for CI/CD, and implemented Express rate limiting to mitigate resource abuse, sustaining **99.5% uptime** over 3 months.
- ***Tech: PostgreSQL, Node.js, Express.js, JavaScript, Prisma, Docker, AWS EC2, Nginx, GitHub Actions***

Daigo | GitHub **May 2026 – Present**

- Building a two-sided marketplace with **role-based auth across 5 tiers** (rider, driver, moderator, admin, super-admin), vehicle verification, and **PostGIS geospatial matching** via Prisma `$queryRaw` for detour limits and **multi-hop relay matching** (chaining two drivers).
- Engineered **first-accept-wins concurrency** using **Prisma transactions** and a server-enforced 75-second expiry; routed cross-instance events over **WebSocket + Redis pub/sub** for real-time driver alerts and in-trip chat.
- Implemented **failure-resistant backend patterns**: idempotency keys for payment retries, Razorpay webhook verification, atomic seat release on cancellation, and **revocable JWTs with refresh token rotation**.
- Built a **dynamic pricing engine** (zone, time-of-day, demand multipliers) with two hard caps checked against the **rounded** value (eliminating floating-point pricing bugs); offloaded heavy tasks (exports, alerts) to **Upstash QStash**.
- Integrated **signed-URL direct-to-storage uploads** (Supabase) — validating file type/size server-side before clients upload directly, drastically reducing API memory footprint.
- ***Tech: Next.js, TypeScript, Prisma, PostgreSQL + PostGIS, Redis, QStash, Stadia Maps, Razorpay***

Achievements

- Selected for **GirlScript Summer of Code** 2026.
- Winner of **CodeaThon 7.0** (cash prize, 120+ participants).
- **313 LeetCode** problems solved | **20+ GitHub repos** | **1000+ commits**.
- Shortlisted for **Round 2 – Polaris Fellowship 2026** (selected from thousands across 300+ cities).
- **CEED Member** (Entrepreneurship Program), Chitkara University.

Certifications

- **AWS Educate Introduction to Cloud 101** – EC2, S3, IAM fundamentals
- **Web Development Masterclass** – HTML/CSS, JavaScript, Node.js, REST APIs

Education

Chitkara University **Aug 2024 – Expected 2027**

Bachelor of Computer Applications (BCA) | CGPA: 9.44/10 *Himachal Pradesh, India*

- Coursework: DSA, Operating Systems, DBMS, Software Engineering, Computer Networking, Web Performance